Executive Summary – House Price Prediction

Project Objective

The goal of this project was to develop a regression model capable of predicting house prices using property-related features and advanced preprocessing techniques.

Dataset

The dataset contains house characteristics such as area, bedrooms, bathrooms, location, and other relevant housing attributes used to estimate property prices.

Data Preprocessing

Several preprocessing techniques were applied:

- Missing value imputation
- One-hot encoding for categorical variables
- Feature scaling
- Log transformation of the target variable

Models Evaluated

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Evaluation Metrics

Performance was measured using:

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 Score

Results

Ensemble models such as Random Forest and Gradient Boosting generally outperformed Linear Regression due to their ability to capture complex relationships within the data.

Residual Analysis

Residual plots showed prediction errors were centered around zero, indicating that the selected model generalized well on unseen data.

Conclusion

The project successfully implemented a complete regression workflow including preprocessing, feature engineering, model comparison, and prediction. The best-performing model was saved for future deployment and inference.